

q. 4

Part A

Determine the thermal voltage (V_T) at 25°C

We use the formula $V_T = \frac{kT}{q}$

1. $T_K = T_C + 273^\circ \rightarrow T = 25 + 273 = 298\text{K}$

2. identify constants: * Boltzmann's constant k
 $= 1.38 \times 10^{-23} \text{ J/K}$

• Charge of an electron $q \approx 1.602 \times 10^{-19} \text{ C}$

3. calculate:

$$V_T = \frac{(1.38 \times 10^{-23}) \times 298}{1.602 \times 10^{-19}}$$

$$V_T = 0.02567 \text{ V} / 25.7 \text{ mV}$$

Part B: Find the diode current (I_D)

1. calculate the exponent first

$$\frac{V_D}{nV_T} = \frac{0.5}{2 \times 0.02567} \approx 9.739$$

2. Plug into the main equation:

$$I_D = (40 \times 10^{-9}) \times (e^{9.739} - 1)$$

$$I_D = (40 \times 10^{-9}) \times (16966.5 - 1)$$

$$I_D = 0.000678 \text{ A}$$

$$\therefore I_D = 0.678 \text{ mA}$$